LAB 6:

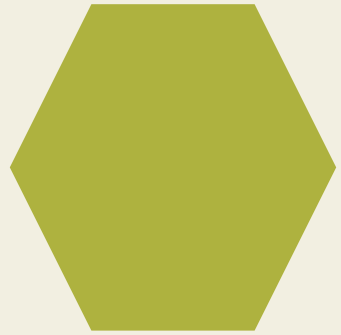
Quantitative Reasoning I

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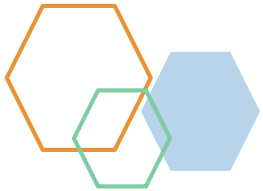
Introduction to Engineering

Fall 2026



TODAY'S LAB · 80 MINUTES

Lab Agenda





Recap of Lecture 6

Demand and capacity, utilisation, rates and flows, peaks, order-of-magnitude checks.

15 min



Instructions for Group Work

The nine-step quantitative task, and the case scenario your team receives.

10 min



Group Work — Quantify Your System

Process flow, utilisation, bottleneck, flows, buffer, peak, improvement, reasonableness check.

35 min · evaluated



Worked Example: Case #1

Semiconductor Wafer Packaging Line, solved step by step.

20 min

RECAP · LECTURE 6

Demand, Capacity and Utilisation

DEFINITION

Demand = what a system is asked to deliver. **Capacity** = the most it can deliver. Both need **units and a time basis**.

Utilisation = Demand ÷ Capacity

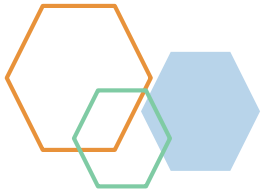
< ≈ 0.8	≈ 0.8 – 1.0	> 1.0
comfortable	***stressed***	***over capacity***
Room to absorb variation.	*Little margin; delays appear.*	*Demand is turned away or a queue grows.*

Lecture example: faculty building 180 ÷ 250 kW = 0.72 · dining hall 360 ÷ 400 per hour = 0.90 · campus bus 300 ÷ 240 per hour = 1.25.

Always compare like with like: same unit, same time basis, before you divide.

RECAP · LECTURE 6

Rates, Flows and Stocks



RATE

Quantity per unit time

km/h, kW = kWh/h, units/h, units/day.
kW is a rate; kWh is a quantity.
Power × time = energy.
Carry units in [square brackets].

STOCK AND FLOW

Change = inflow − outflow

A stock (queue, buffer, battery) grows when in > out and drains when out > in.
Time to fill = free space ÷ (in − out).

TWO FLOW RULES

Steady-state shortcuts

Flow = speed × density.
Number inside = arrival rate × time inside.
Both hold for steady conditions.

Average hides the peak

Faculty building: average 94 kW, peak 180 kW at 13:00, transformer 250 kW. A system sized for the average would fail at midday — size for the peak.

Change in stock = flow in − flow out. If it is positive, something is building up.

RECAP · LECTURE 6

Closing a Gap and Checking the Scale

Three ways to close a demand–capacity gap

OPTION 1

Reduce or shift demand

e.g. stagger charging in two batches.

OPTION 2

Raise capacity

A bigger connection or faster machine. Costs money and time.

OPTION 3

Accept waiting

Let a queue form; decide how long a wait is tolerable.

Order of magnitude: estimate before you calculate

1 Define the quantity and its units

2 Break it into factors you can guess

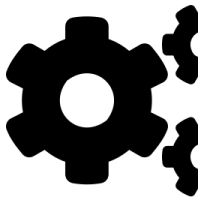
3 Round each factor to one digit, multiply

4 Check units; compare with a known value

Unit traps: kWh ≠ kW · MW → kW is ×1,000 · 36 min = 0.6 h, not 0.36 h · no number without unit and time basis.

Most estimation errors are unit or counting errors, not hard mathematics.

GROUP WORK · EVALUATED · 40 MINUTES

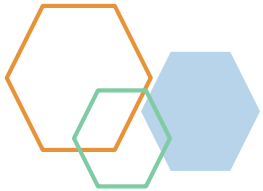


Quantify the System

Different case scenario exercises.

GROUP WORK · LAB 6 · STEPS 1–5

Your Task: Quantify the System (1/2)

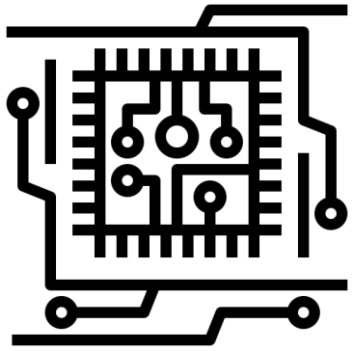


1. Draw the system as a 4–6 stage process flow.
2. Identify demand and capacity, with units and time basis.
3. Calculate utilisation at the important stages: Utilisation = Demand ÷ Capacity.
4. Classify each key stage: comfortable (< ≈0.8), stressed (≈0.8–1.0), or over capacity (> 1.0).
5. Identify the bottleneck and state the maximum practical system throughput.

GROUP WORK · LAB 6 · STEPS 6–10

Your Task: Quantify the System (2/2)

6	Calculate the material/unit inflow and outflow through the process, including rejects, losses, rework or scrap.
7	Calculate **one queue**, buffer or stock change using Change = inflow − outflow; find its size after a stated time, or when it becomes full.
8	Compare average demand with peak demand; explain whether the system performs differently at peak.
9	Propose one quantified improvement (reduce/shift demand, increase capacity, or accept/manage waiting). Show the new utilisation or capacity.
10	Order-of-magnitude check on one answer: state the expected scale, check units, say whether the result is plausible.



Semiconductor Wafer Packaging Line

WORKED EXAMPLE · CASE #1

Semiconductor Wafer Packaging Line

1 Semiconductor Wafer Packaging Line

A semiconductor plant packages power-control chips for industrial electronics. During a normal shift, 7,200 dies/day are released from wafer fabrication. The line operates 12 h/day. The product starts as a silicon wafer. The chips on it are first sorted (die sort). The good chips are fixed onto a carrier (die attach) and connected to their pins with very fine wires (wire bonding). Each assembled chip is then electrically tested. Last, the chips are packaged for shipping. Die Sort can process 680 dies/h, Die Attach 620 dies/h, Wire Bonding 580 dies/h, Electrical Test 600 dies/h, and Final Packaging 640 units/h. Historical data show that 2% of dies are rejected after Die Sort, 3% are lost during Die Attach/Wire Bonding, and 4% fail Electrical Test. Failed electrical-test units are scrapped.



Average daily demand is 6,000 packaged chips/day, but during the final three days of each month demand rises to 7,000/day.

Before Wire Bonding there is a buffer that currently contains 300 units and can hold at most 1,100 units. During one observed peak hour, Die Attach sent 600 units/h into this buffer while Wire Bonding processed only 560 units/h.

Capacities (units per hour)

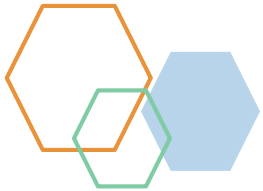
- Die Sort: 680 dies/h
- Die Attach: 620 dies/h
- Wire Bonding: 580 dies/h
- Electrical Test: 600 dies/h
- Final Packaging: 640 units/h

Rejection rates

- 2% of dies are rejected after Die Sort
- 3% are lost during Die Attach/ Wire Bonding
- 4% fail Electrical Test (scrapped)

WORKED EXAMPLE · CASE #1

Semiconductor Wafer Packaging Line



THE CASE

A plant packages power-control chips for industrial electronics. During a normal shift, 7,200 dies/day are released from wafer fabrication. The line operates 12 h/day. Dies are sorted (Die Sort), fixed onto a carrier (Die Attach), wired to pins (Wire Bonding), electrically tested, and packaged for shipping.

CAPACITY (dies or units / h)

Stage rates

Die Sort 680
Die Attach 620
Wire Bonding 580
Electrical Test 600
Final Packaging 640

LOSSES AND DEMAND

Yield and demand

2% rejected after Die Sort
3% lost in Die Attach/Wire Bonding
4% fail Electrical Test (scrapped)
Demand: 6,000/day average; 7,000/day in the last 3 days of each month

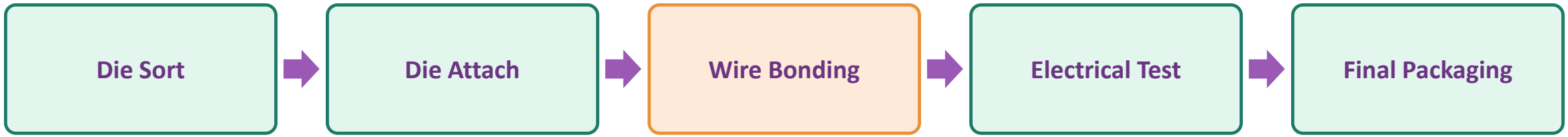
BUFFER BEFORE WIRE BONDING

Peak-hour observation

Holds 300 now, max 1,100 units.
Die Attach sent 600 units/h in;
Wire Bonding processed 560 units/h.

SOLUTION · STEPS 1–2

Process Flow and Daily Capacity



Capacity per day = rate [per h] × 12 [h/day]

Stage	Rate [chips/ h]	Time [h/day]	Capacity [per day]
Die Sort	680	12	**8,160 = 680*12**
Die Attach	620	12	7,440
Wire Bonding	580	12	6,960
Electrical Test	600	12	7,200
Final Packaging	640	12	7,680

Demand: 6,000 chips/day (average), 7,000 chips/day (peak). Both are per day, so we compare with capacity per day.

SOLUTION · STEPS 3–5

Utilisation, Classification and Bottleneck

Stage	Capacity [/day]	Util. at 6,000	Util. at 7,000	Judgement at peak
Die Sort	8,160	**0.74**	**0.86**	***stressed***
Die Attach	7,440	**0.81**	**0.94**	***stressed***
Wire Bonding	6,960	**0.86**	**1.01**	***over capacity***
Electrical Test	7,200	**0.83**	**0.97**	***stressed***
Final Packaging	7,680	**0.78**	**0.91**	***stressed***

Utilisation = demand [chips/day] ÷ capacity [chips/day]; e.g. Wire Bonding: 6,000 ÷ 6,960 = 0.86 and 7,000 ÷ 6,960 = 1.01.

BOTTLENECK

Wire Bonding: 6,960 per day

The lowest capacity sets the maximum practical throughput of the whole line.

READING THE TABLE

At average demand every stage is below 1.0. At peak, four stages are stressed and Wire Bonding is over capacity.

SOLUTION · STEPS 6–7

Material Flow and Buffer Build-up

Good chips from 7,200 dies per day

7,200
dies released

→

7,056
after Die Sort
7,200 × 0.98

→

6,844
after attach/bond
7,056 × 0.97

→

≈ 6,571
good chips after test
6,844 × 0.96

Losses per day: 144 rejected at Die Sort, ≈212 lost in attach/bond, ≈273 scrapped at test; about 629 of 7,200 dies (≈ 9%) never ship.

BUFFER BEFORE WIRE BONDING · PEAK HOUR

Change = in − out = 600 − 560 = 40 units/h

Free space = 1,100 − 300 = 800 units

Time to fill = 800 ÷ 40 = 20 h

A 12 h shift at this rate adds 480 units (reaching 780 of 1,100); the buffer would be full after 20 h, so Die Attach would then be blocked.

SOLUTION · STEPS 8–9

Average vs Peak, and One Improvement

AVERAGE · 6,000 / DAY

System can cope

Capacity of the bottleneck is 6,960, so utilisation is 0.86. Wire Bonding is stressed, but no stage is over capacity.

PEAK · 7,000 / DAY

Slightly over capacity

Wire Bonding at 1.01 cannot quite keep up; the other stages are at 0.86–0.97, so any disturbance builds a queue.

BEHAVIOUR

It differs at peak

At average demand the line has some margin. At peak the bottleneck saturates, and the buffer fills as in the previous step.

QUANTIFIED IMPROVEMENT · INCREASE CAPACITY AT WIRE BONDING

Target: peak utilisation ≤ 0.85

Required capacity = 7,000 ÷ 0.85 ≈ 8,235 per day = 8,235 ÷ 12 ≈ 686 per h

Current 580 per h → increase by (686 − 580) ÷ 580 ≈ 18%

New utilisation at peak: 7,000 ÷ 8,235 = 0.85. Test (7,200 per day) then becomes the next-tightest stage and should be checked.

SOLUTION · STEP 10

Order-of-Magnitude Check

1 · EXPECTED SCALE

Thousands per day

About 7,000 dies in, a few percent lost at each of three stages: output should be a little below 7,000, i.e. thousands per day — not tens, not millions.

2 · UNITS

Consistent

[dies/day] × [fraction kept, no unit] = [chips/day]. Hours: [chips/h] × [h/day] = [chips/day]. Same time basis on both sides.

3 · KNOWN VALUE

Plausible

Rough check: 7,000 × 0.9 ≈ 6,300–6,600. Our result, ≈ 6,571 chips/day, sits inside the expected 6,500–7,000 range.

Verdict: 6,500–7,000 chips/day is plausible. A result like 65 or 6,500,000 would signal a unit or power-of-ten slip.

Note: output (≈ 6,571) is above the average demand of 6,000 and below the peak demand of 7,000: demand is met on average, and only the peak days fall short.

Thank You — Questions?

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